

Ion- π Interactions: Empirical formulation involving 12-6-4 Lennard-Jones Potential

Progress Report in Partial Fulfilment of the Requirements for the
Chemistry Minor Project (CHY 450)

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DECLARATION

I, Sougat Dasgupta (IMS22299), hereby declare that the progress report on the research work entitled “Ion- π Interactions: Empirical formulation involving 12-6-4 Lennard-Jones Potential” submitted to School of Chemistry, Indian Institute of Science Education and Research Thiruvananthapuram, towards partial requirement for the Chemistry Minor Project (CHY 450) is an original work carried out by me under the supervision of Prof. R. S. Swathi.

In keeping with the general practise of reporting scientific observations, due references have been made in the report. Any omission that might have occurred by oversight or error is regretted.

Thiruvananthapuram

Date: 03/05/2026



Sougat Dasgupta

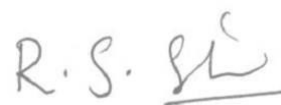
Signature

Certificate

This is to certify that the research work contained in this progress report entitled “Ion- π Interactions: Empirical formulation involving 12-6-4 Lennard-Jones Potential” submitted by Sougat Dasgupta (IMS22299) to Indian Institute of Science Education and Research Thiruvananthapuram towards partial requirement for the Chemistry Minor Project (CHY 450) is an original work carried under my supervision and has not been submitted elsewhere for the award of any degree.

Thiruvananthapuram

Date: 03/05/2026



Prof. R. S. Swathi

Signature (with date and official seal)

Research Abstract

Ion- π interactions, the non-covalent forces between charged species and π systems, play a fundamental role in chemical and biological research. Although there are studies in the literature for systems like cations with pristine π -systems and anions with fluorinated π -systems, complementary systems like cations with fluorinated π -systems and anions with pristine π -systems have not been explored much.

In this work, we develop empirical force field parameters with 12-6-4 Lennard-Jones potential augmented with explicit Coulomb electrostatics, fitted to high-level ab initio reference data. The adopted methodology provides a computationally efficient alternative to ab initio methods for the evaluation of ion- π interaction energetics, and represent a step toward force field development for molecular dynamics simulations.

Research Overview

An ion- π interaction is a strong non-covalent force between a charged ion and the polarizable electron cloud of a π system, such as a double bond or an aromatic ring. In a cation- π interaction, a positively charged ion is directly attracted to the negatively charged, electron-rich face of the π cloud. Conversely, an anion- π interaction involves a negatively charged ion. These interactions are fundamental to chemical and biological landscapes, playing critical roles in protein folding, enzyme catalysis, and the design of sophisticated supramolecular hosts^[1,2] and have direct implications in molecular dynamics simulations. Although cations interacting with pristine π -systems and anions interacting with fluorinated π -systems are well-studied, there is a significant lack of literature regarding complementary combinations: cations with fluorinated π -systems and anions with pristine π -systems. In these systems, the interaction often leads to metastable states or stable bound states on the potential energy surface. Standard force fields often fail to describe these transition regimes accurately, leaving a gap in our ability to simulate complex molecular environments where such seemingly unfavorable interactions persist^[3].

The governing factor in these interactions is the molecular quadrupole moment (Q_{zz}), where z denotes the direction perpendicular to the molecular plane, defining the deviation of a charge

distribution from spherical symmetry. In pristine hydrocarbon π -systems, the low electronegativity of the hydrogen atoms results in a concentrated, electron-rich π -cloud above and below the molecular surface, characterized by a negative quadrupole moment ($Q_{zz} < 0$). However, upon fluorination, the highly electronegative fluorine substituents exert a strong inductive effect, pulling electron density toward the molecular periphery. This creates a localized, electropositive region at the center of the π -system, effectively reversing the sign of the quadrupole moment ($Q_{zz} > 0$)^[4,5].

The standard Lennard-Jones (LJ) potential, widely used in classical force fields for non-covalent interactions, fails to capture the intricate features of intermolecular interactions owing to limitations in both its short-range repulsive wall and its long-range attractive decay^[2,6].

In this study, our aim is to develop and validate 12-6-4 LJ parameters augmented with explicit electrostatic terms for ion- π interactions in a systematically chosen series of pristine and fluorinated π -systems. The standard 12-6 LJ potential fails to reproduce the correct interaction energetics for highly polarizable ion- π complexes because it neglects the contribution from induction^[2,7,8].

The 12-6-4 LJ model accounts for the ion-induced dipole interaction by adding a C_4/r^4 polarization term, and has been shown to significantly improve the description of systems where induction is dominant^[8,9]. The specific objectives are: (i) To utilize high-level *ab initio* potential energy curves (computed at the CCSD(T) and DLPNO-CCSD(T) levels) for a set of ion- π complexes involving pristine and fluorinated π -systems (C_6H_6 , C_4H_4 , C_2H_4 , C_6F_6 , C_4F_4 , C_2F_4) with monovalent cations (Li^+ , Na^+ , K^+) and anions (F^- , Cl^- , Br^-); (ii) To parametrize the 12-6-4 LJ potential against reference coupled-cluster data.

Results and Discussion

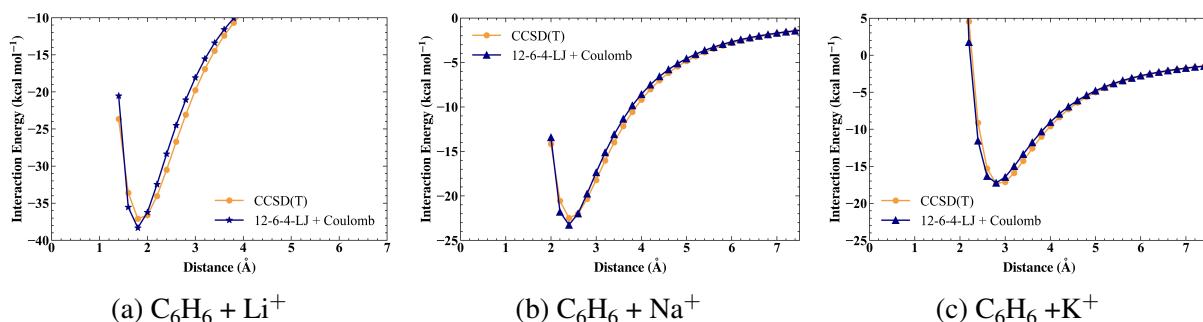


Figure 1: Comparison of energy profiles for the interaction of Li⁺, Na⁺, and K⁺ with C₆H₆ using 12-6-4 LJ + Coulomb vs CCSD(T).

Cations	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_H (kcal mol ⁻¹)	$r_{0,H}$ (Å)	$C_{4,H}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
Li ⁺	0.002	3.79	5.0	0.94	3.17	89.75	0.935
Na ⁺	0.01	3.99	10.03	0.87	3.45	20.01	0.440
K ⁺	0.01	4.24	10.01	0.57	3.93	20.03	0.787

Table 1: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-hydrogen interactions in C₆H₆-Cation systems

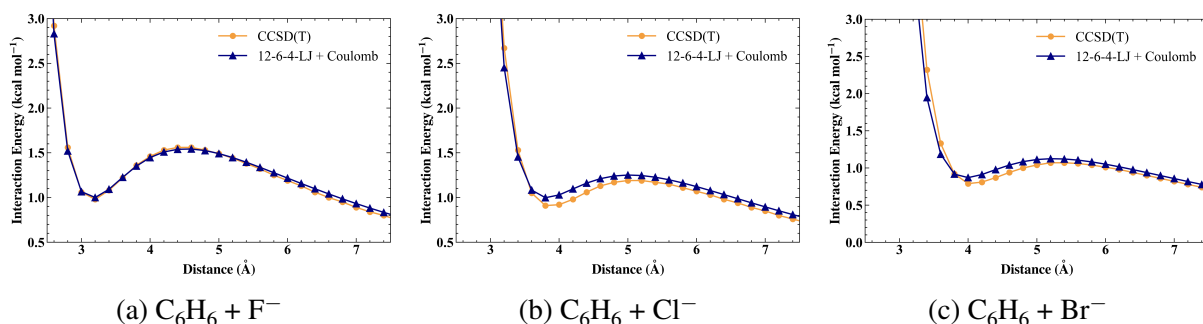


Figure 2: Comparison of energy profiles for the interaction of F⁻, Cl⁻, and Br⁻ with C₆H₆ using 12-6-4 LJ + Coulomb vs CCSD(T).

As seen in Fig[1], the interaction between benzene and cations (Li⁺, Na⁺, K⁺) is characterized by a single stable minima and a tail. This is the classic cation- π interaction where the electron-rich π -cloud of the hydrocarbon ring provides an attractive environment for the positive charge of the incoming cation. In contrast, benzene's interaction with anions as in Fig[2] exhibits a local minimum with a repulsion barrier and then a tail. This occurs because the negative charge of the anion is repelled by the high electron density of the aromatic π -system, requiring the C_4 induction term to work against a strong electrostatic repulsion. The behavior

Anions	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_H (kcal mol ⁻¹)	$r_{0,H}$ (Å)	$C_{4,H}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
F ⁻	0.01	3.80	37.19	0.98	3.66	163.93	0.495
Cl ⁻	0.017	4.50	30.02	0.86	3.95	155.02	0.091
Br ⁻	0.02	4.64	149.99	0.47	4.25	50.00	0.207

Table 2: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-hydrogen interactions in C₆H₆-Anion systems

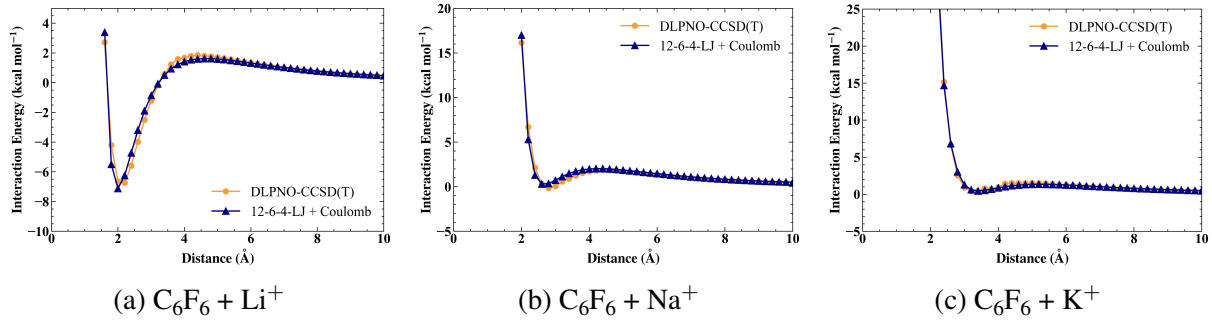


Figure 3: Comparison of energy profiles for the interaction of Li⁺, Na⁺, and K⁺ with C₆F₆ using 12-6-4 LJ + Coulomb vs DLPNO-CCSD(T).

Cations	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_F (kcal mol ⁻¹)	$r_{0,F}$ (Å)	$C_{4,F}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
Li ⁺	0.002	3.82	15.20	0.12	3.10	25.40	0.312
Na ⁺	0.010	4.10	12.50	0.10	3.35	18.20	0.155
K ⁺	0.010	4.35	11.80	0.07	3.75	15.50	0.210

Table 3: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-fluorine interactions in C₆F₆-Cation systems

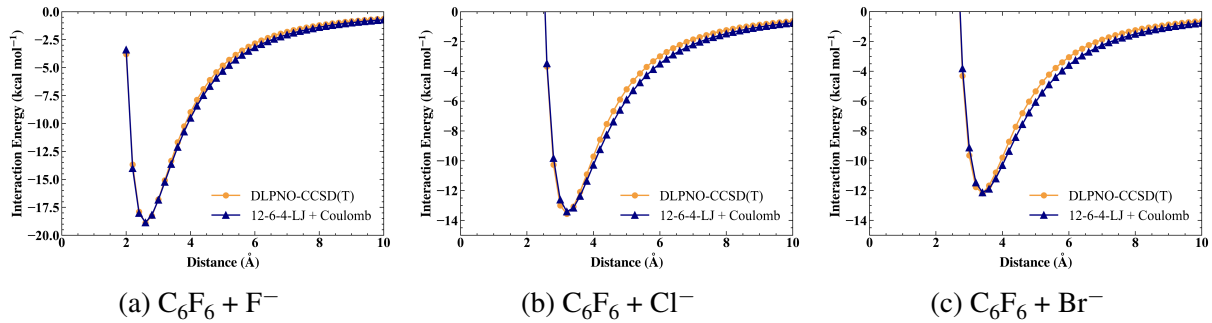


Figure 4: Comparison of energy profiles for the interaction of F⁻, Cl⁻, and Br⁻ with C₆F₆ using 12-6-4 LJ + Coulomb vs DLPNO-CCSD(T).

Anions	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_F (kcal mol ⁻¹)	$r_{0,F}$ (Å)	$C_{4,F}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
F ⁻	0.015	4.25	45.10	0.08	3.50	90.50	0.420
Cl ⁻	0.020	4.80	40.00	0.06	4.10	85.00	0.180
Br ⁻	0.025	5.00	130.0	0.04	4.40	40.00	0.250

Table 4: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-fluorine interactions in C₆F₆-Anion systems

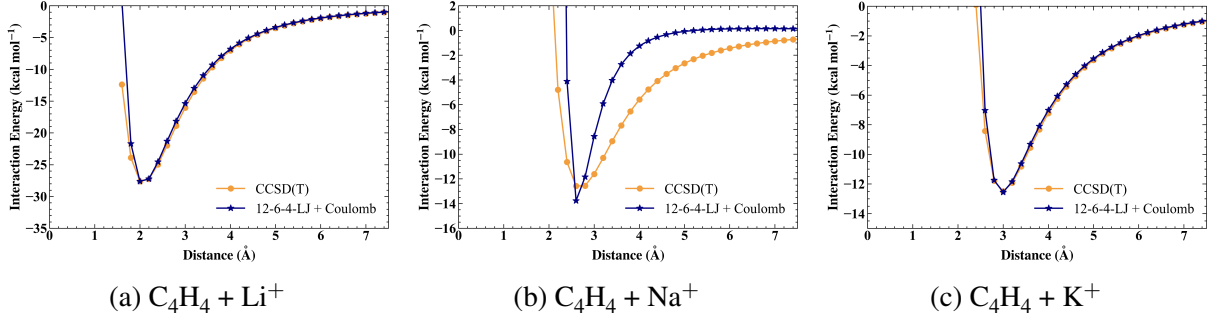


Figure 5: Comparison of energy profiles for the interaction of Li⁺, Na⁺, and K⁺ with C₄H₄ using 12-6-4 LJ + Coulomb vs CCSD(T).

Cations	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_H (kcal mol ⁻¹)	$r_{0,H}$ (Å)	$C_{4,H}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
Li ⁺	0.005	3.45	5.0	1.30	3.20	143.97	2.062
Na ⁺	0.015	3.67	5.0	0.72	3.59	119.62	1.526
K ⁺	0.050	3.90	5.0	0.50	3.85	147.47	1.198

Table 5: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-hydrogen interactions in C₄H₄-Cation systems

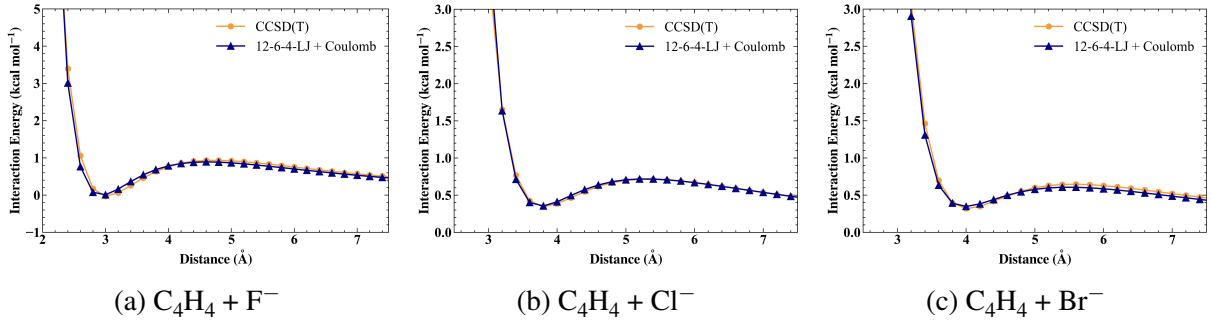


Figure 6: Comparison of energy profiles for the interaction of F⁻, Cl⁻, and Br⁻ with C₄H₄ using 12-6-4 LJ + Coulomb vs CCSD(T).

Anions	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_H (kcal mol ⁻¹)	$r_{0,H}$ (Å)	$C_{4,H}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
F ⁻	0.01	3.70	100.00	0.58	3.56	97.37	0.104
Cl ⁻	0.01	4.44	113.23	0.64	4.01	20.00	0.057
Br ⁻	0.01	4.54	142.14	0.39	4.27	34.18	0.058

Table 6: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-hydrogen interactions in C₄H₄-Anion systems

Cations	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_F (kcal mol ⁻¹)	$r_{0,F}$ (Å)	$C_{4,F}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
Li ⁺	0.630	1.84	40.46	0.34	1.67	10.00	0.881
Na ⁺	0.001	3.20	15.00	0.01	2.86	15.00	0.855
K ⁺	0.001	3.37	15.00	0.01	3.23	25.00	0.771

Table 7: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-fluorine interactions in C₄F₄-Cation systems.

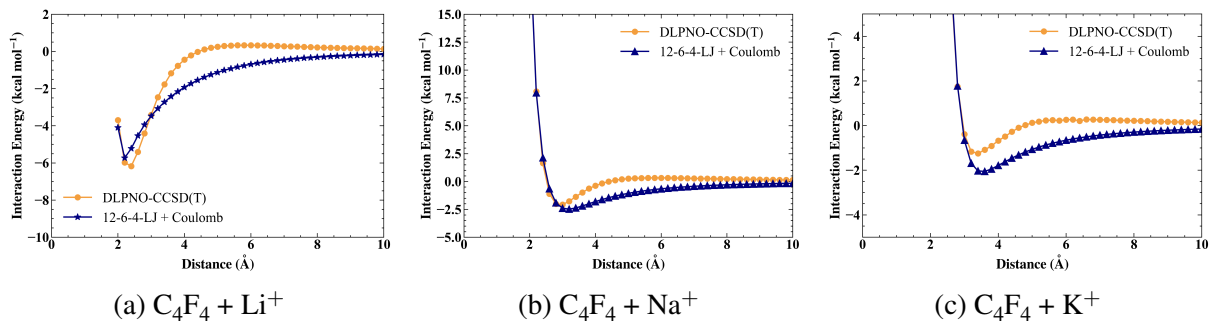


Figure 7: Comparison of energy profiles for the interaction of Li^+ , Na^+ , and K^+ with C_4F_4 using 12-6-4 LJ + Coulomb vs DLPNO-CCSD(T).

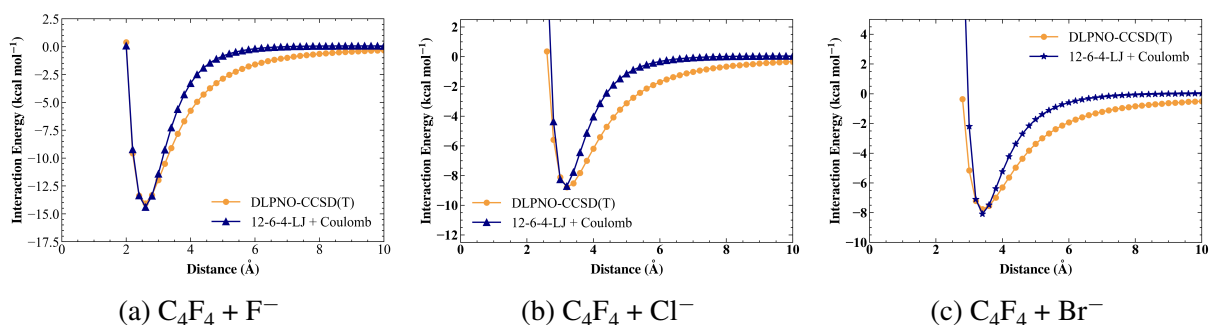


Figure 8: Comparison of energy profiles for the interaction of F^- , Cl^- , and Br^- with C_4F_4 using 12-6-4 LJ + Coulomb vs DLPNO-CCSD(T).

Anions	ε_C (kcal mol $^{-1}$)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol $^{-1}$ Å 4)	ε_F (kcal mol $^{-1}$)	$r_{0,F}$ (Å)	$C_{4,F}$ (kcal mol $^{-1}$ Å 4)	RMSD (kcal mol $^{-1}$)
F^-	0.301	2.01	62.73	0.005	3.48	97.32	1.345
Cl^-	0.066	2.24	97.56	0.083	3.05	96.06	1.601
Br^-	0.005	2.30	118.70	0.005	4.09	144.46	2.201

Table 8: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-fluorine interactions in C_4H_4 -Anion systems.

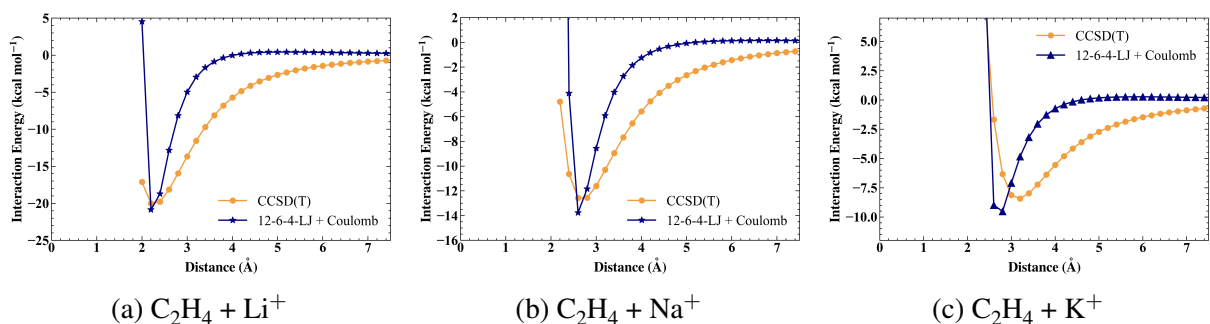


Figure 9: Comparison of energy profiles for the interaction of Li^+ , Na^+ , and K^+ with C_2H_4 using 12-6-4 LJ + Coulomb vs CCSD(T).

Cations	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_H (kcal mol ⁻¹)	$r_{0,H}$ (Å)	$C_{4,H}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
Li ⁺	0.001	2.30	39.26	0.05	1.99	54.13	4.113
Na ⁺	0.100	2.00	293.83	0.16	2.20	5.00	10.830
K ⁺	0.030	3.14	30.11	0.01	1.00	80.22	2.730

Table 9: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-hydrogen interactions in C₂H₄-Cation systems.

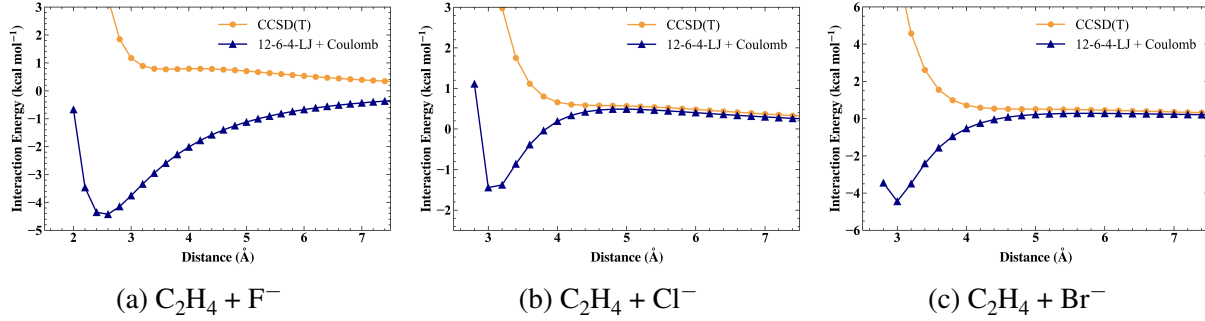


Figure 10: Comparison of energy profiles for the interaction of F⁻, Cl⁻, and Br⁻ with C₂H₄ using 12-6-4 LJ + Coulomb vs CCSD(T).

Anions	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_H (kcal mol ⁻¹)	$r_{0,H}$ (Å)	$C_{4,H}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
F ⁻	0.001	2.50	2.00	0.001	1.50	5.00	5.290
Cl ⁻	0.017	3.60	30.00	0.001	2.40	50.00	1.973
Br ⁻	0.020	3.50	100.00	0.001	3.00	50.00	3.947

Table 10: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-hydrogen interactions in C₂H₄-Anion systems.

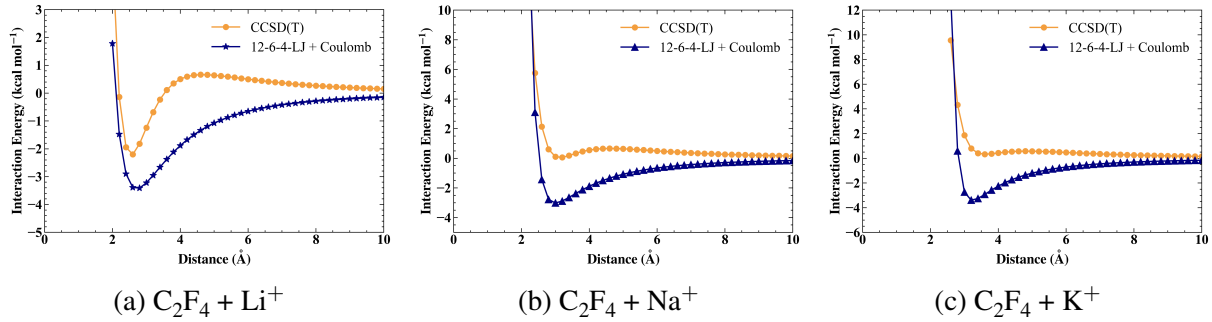


Figure 11: Comparison of energy profiles for the interaction of Li⁺, Na⁺, and K⁺ with C₂F₄ using 12-6-4 LJ + Coulomb vs CCSD(T).

Cations	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_F (kcal mol ⁻¹)	$r_{0,F}$ (Å)	$C_{4,F}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
Li ⁺	0.001	0.80	2.00	0.001	1.00	2.00	1.460
Na ⁺	0.001	0.82	2.00	0.001	2.92	2.00	1.757
K ⁺	0.002	3.80	10.00	0.010	2.83	10.00	1.951

Table 11: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-fluorine interactions in C₂F₄-Cation systems.

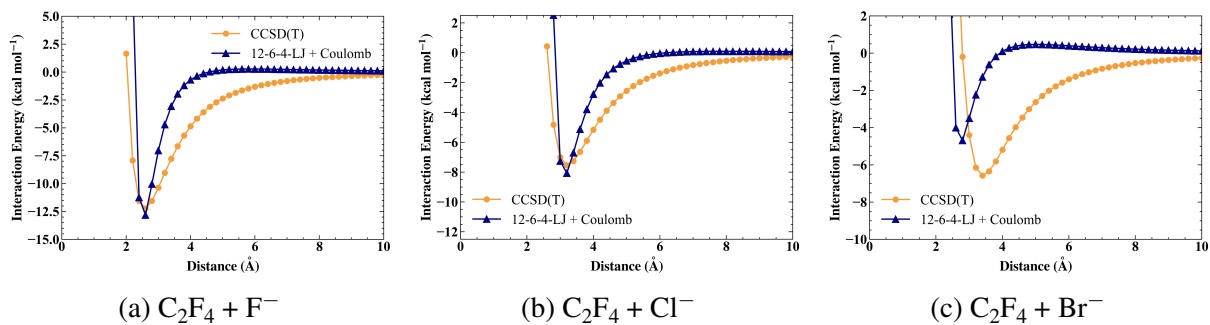


Figure 12: Comparison of energy profiles for the interaction of F⁻, Cl⁻, and Br⁻ with C₂F₄ using 12-6-4 LJ + Coulomb vs CCSD(T).

Anions	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	$C_{4,C}$ (kcal mol ⁻¹ Å ⁴)	ε_F (kcal mol ⁻¹)	$r_{0,F}$ (Å)	$C_{4,F}$ (kcal mol ⁻¹ Å ⁴)	RMSD (kcal mol ⁻¹)
F ⁻	0.50	1.10	78.64	0.500	2.32	50.00	3.345
Cl ⁻	0.05	2.24	102.32	0.005	3.47	108.39	3.237
Br ⁻	0.01	2.60	29.02	0.010	2.70	41.85	7.480

Table 12: Optimal 12-6-4 LJ + Coulomb parameters for ion-carbon and ion-fluorine interactions in C₂H₄-Anion systems.

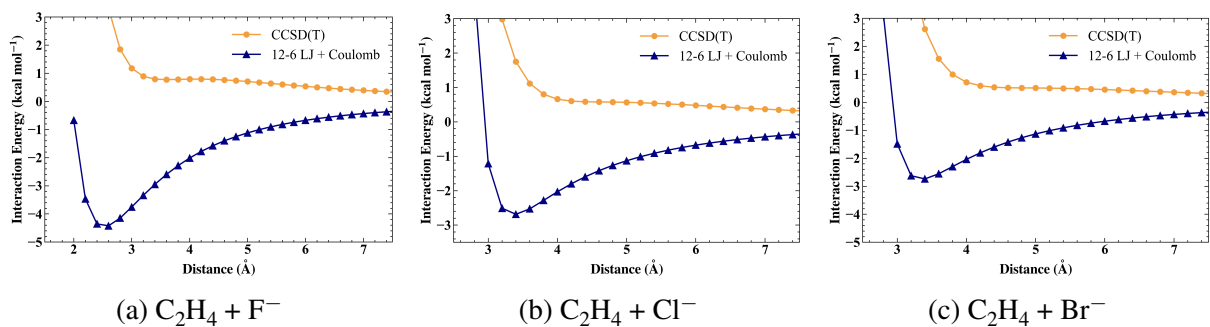


Figure 13: Comparison of energy profiles for the interaction of F⁻, Cl⁻, and Br⁻ with C₂H₄ using 12-6 LJ + Coulomb vs CCSD(T).

Anions	ε_C (kcal mol ⁻¹)	$r_{0,C}$ (Å)	ε_H (kcal mol ⁻¹)	$r_{0,H}$ (Å)	RMSD (kcal mol ⁻¹)
F ⁻	0.001	2.50	0.001	1.50	5.290
Cl ⁻	0.017	3.60	0.001	2.40	2.276
Br ⁻	0.020	3.50	0.001	3.00	3.947

Table 13: Optimal 12-6 LJ + Coulomb parameters for ion-carbon and ion-hydrogen interactions in C₂H₄-Anion systems.

is fundamentally reversed for hexafluorobenzene. Because fluorine is the most electronegative element, the six F-atoms withdraw significant electron density from the ring creating positive charge on the face of the ring. For C_6F_6 with cations as in Fig[3], we observe a significant repulsion barrier. The ring has become electron-deficient, and the cations now encounter a repulsive electrostatic environment that was not present in benzene. This often leads to "metastable" states where the ion is pushed away from the ring center. Conversely, the electron-poor nature of C_6F_6 makes it much more favorable for anions as in Fig[4]. Here, we see stable minima, as the electrostatic term is attractive in nature and induction/polarization effects can also contribute.

The reversal of the nature of the interaction which is observed between benzene (C_6H_6) and hexafluorobenzene (C_6F_6) is also observed in smaller cyclic systems. For cations interacting with pristine π -systems, the interaction strength follows the order $Li^+ > Na^+ > K^+$, and for anions interacting with fluorinated π -systems the interacting strength follows the order $F^- > Cl^- > Br^-$. To check the efficiency of 12-6 LJ potential we tried to parametrize 12-6 LJ against CCSD(T) data for the interaction of ethene(C_2H_4) with anions, but we got very poor fit and high RMSD values which suggests that the C_4 term is essential to account for the induction effect. As 12-6 had limitations, we tried 12-6-4 LJ + coulomb model to parametrize it against our coupled-cluster data. We got good RMSD fits for all the data except the fit for C_2H_4 - cations and C_2H_4 - anions which were the most significant challenges that we encountered while fitting. Despite our best efforts, the best results we could get so far is shown in Fig[11, 12] but they remain unsatisfactory. This implies that 12-6-4 LJ + coulomb model is able to capture the energetics of the ion- π interactions reasonably well. The discrepancies observed for the C_2H_6 - cation and C_2H_6 - anion interaction might be due to the deficiencies of the method chosen. In our model, we do not consider anisotropic nature of the ion- π interactions which may be crucial in characterizing these interactions.

Theoretical Details

This study consists of three pristine π -systems: ethene (C_2H_4), cyclobutadiene (C_4H_4), and benzene (C_6H_6) and their fluorinated counterparts: perfluoroethene (C_2F_4), perfluorocyclobutadiene (C_4F_4), and perfluorobenzene (C_6F_6). The ions considered are three monovalent alkali metal cations (Li^+ , Na^+ , K^+) and three monovalent halide anions (F^- , Cl^- , Br^-), yielding a total of 36 ion- π system combinations. In each system, the ion is placed along the axis perpendicular to the molecular plane and allowed to pass through the ring centroid. The interaction energy is evaluated as a function of the ion-centroid distance r . The equilibrium geometries of the π -systems were obtained at the M06-2X/cc-pVTZ level of theory, with the exception of benzene, for which the highly accurate coupled-cluster equilibrium structure reported by Gauss and Stanton^[5] was employed. High-level coupled-cluster interaction energy profiles were provided by prior studies in our group as reference data for the parametrization^[13]. For the pristine π -systems (C_2H_4 , C_4H_4 , C_6H_6) interacting with all six ions, and for perfluoroethene (C_2F_4) with all ions, the interaction energies were computed at the CCSD(T)/aug-cc-pVTZ level of theory using the Gaussian software package^[12]. For the fluorinated π -systems (C_4F_4 and C_6F_6) interacting with halide anions, where the larger molecular size and the heavier atoms make canonical CCSD(T) computationally expensive, the DLPNO-CCSD(T) method^[11] was employed with the aug-cc-pVTZ basis set, using the ORCA software package. For the K^+ systems, the scalar-relativistic aug-cc-pVTZ-X2C basis set was used to account for relativistic effects. All interaction energies were counterpoise-corrected for basis set superposition error using the Boys-Bernardi procedure. The resulting 36 potential energy profiles constitute the benchmark dataset against which the empirical potential is parameterized.

Empirical Potential Model

The total interaction energy between an ion and a π -system is modeled as a sum of pairwise atom-ion contributions, each described by a 12-6-4 Lennard-Jones potential augmented with a Coulomb electrostatic term^[7,9]:

$$E_{\text{AP}} = \sum_{j=1}^N [V(r_{ij}) + V_{\text{EL}}(r_{ij})] \quad (1)$$

where the sum runs over all N atoms of the π -system, and r_{ij} is the distance between the

ion i and atom j . The 12-6-4 Lennard-Jones term is given by^[6,9]:

$$V(r_{ij}) = \varepsilon \left[\left(\frac{r_{\min}}{r_{ij}} \right)^{12} - 2 \left(\frac{r_{\min}}{r_{ij}} \right)^6 \right] - \frac{C_4}{r_{ij}^4} \quad (2)$$

where ε is the depth of the potential well, $r_{\min}$ is the equilibrium distance at which the pairwise potential reaches its minimum, and C_4 is the polarization coefficient that captures the ion-induced dipole interaction^[9]. The standard 12-6 LJ potential is recovered when $C_4 = 0$. The electrostatic contribution is evaluated using a point-charge Coulomb model:

$$V_{\text{EL}}(r_{ij}) = \frac{C \cdot q_{\text{ion}} \cdot q_j}{r_{ij}} \quad (3)$$

where $C = 332.0636 \text{ kcal mol}^{-1} \text{ \AA} e^{-2}$ is the electrostatic conversion factor, q_{ion} is the charge of the ion ($\pm 1e$), and q_j is the partial charge on atom j of the π -system derived from Merz-Singh-Kollman scheme. The three parameters to be optimized for each ion-atom-type pair are $\{\varepsilon, r_{\min}, C_4\}$. Since the π -systems contain two distinct heavy-atom types (C and H, or C and F), each system involves six parameters in total: $\{\varepsilon_{\text{C}}, r_{0,\text{C}}, C_{4,\text{C}}, \varepsilon_{\text{X}}, r_{0,\text{X}}, C_{4,\text{X}}\}$, where X denotes H or F.

Parametrization Strategy

The optimal parameters were determined by minimizing a weighted root-mean-square deviation (RMSD) between the empirical potential E_{AP} and the ab initio reference energies. The choice of objective function was adapted to the qualitative shape of the reference potential energy curve, which differs fundamentally between two categories of systems.

For systems in which the potential energy curve exhibits a metastable state with a local maximum and a local minimum in addition to a global minimum - as observed for cations with fluorinated- π and anions with pristine- π systems, the objective function takes the form:

$$F_{\text{objective}} = \sqrt{\frac{1}{M} \sum_{i=1}^M \delta_i^2 (e^{-\alpha(x-l_{\max})^2} + e^{-\alpha(x-l_{\min})^2} + e^{-\alpha(x-g_{\min})^2})} \quad (4)$$

where $\delta_i = E_{\text{Ab initio},i} - E_{\text{AP},i}$ is the residual at configuration i , M is the total number of points in the ab initio profile, α is a width parameter controlling the sharpness of the Gaussian weighting, and x is the ion-centroid distance. The coordinates $l_{\max}$, $l_{\min}$, and $g_{\min}$ denote the positions of the local maximum, local minimum, and global minimum of the reference energy

profile, respectively. This formulation assigns enhanced weight to the three critical regions of the curve while suppressing the influence of the long-range tail, where both the absolute energies and the residuals are small.

For systems in which the potential energy curve exhibits only a single well with no local maximum - as observed for cations with pristine- π and anions with fluorinated- π systems, the objective function employs a Boltzmann-type weighting:

$$F_{\text{objective}} = \sqrt{\frac{1}{M} \sum_{i=1}^M \delta_i^2 (e^{-E_i/kT})} \quad (5)$$

where E_i is the ab initio energy at configuration i , k is the Boltzmann constant, and $T = 300$ K, giving $kT = 0.596$ kcal mol⁻¹. The Boltzmann factor concentrates weight near the potential energy minimum, where the interaction is strongest and the physical information is most relevant. In both cases, the minimization was carried out using the Limited-memory Broyden-Fletcher-Goldfarb-Shanno algorithm with bound constraints (L-BFGS-B)^[10]. The L-BFGS-B method is a quasi-Newton local optimizer that approximates the inverse Hessian using a limited-memory BFGS update and handles box constraints on the parameters directly. Physical bounds were imposed on all parameters: $\varepsilon > 0$, $r_{\text{min}} > 0$, and $C_4 \geq 0$, with upper limits guided by values reported in the literature for metal ion-water systems^[8,9].

Conclusions

We analyzed potential energy profiles for the interaction of Li⁺, Na⁺, K⁺, and F⁻, Cl⁻, Br⁻ with C₆H₆, C₄H₄, C₂H₄, C₆F₆, C₄F₄, C₂F₄. Two kinds of interaction energy profiles emerge: attractive systems with a single potential well, and systems with a repulsive barrier followed by metastable states. This is due to the reversal of the quadrupole moment sign, depending on the nature of the system. 12-6-4 Lennard-Jones parameters were developed for ion- π interactions involving pristine and perfluorinated π -systems with monovalent alkali metal cations and halide anions, by fitting against CCSD(T) and DLPNO-CCSD(T) reference interaction energies obtained in our group^[13].

In comparison to the traditional 12-6 LJ + Coulomb model, the 12-6-4 LJ + Coulomb model offers greater robustness in describing ion- π interactions, as it effectively captures both

ion-induction and electrostatic components of the intermolecular interactions.

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